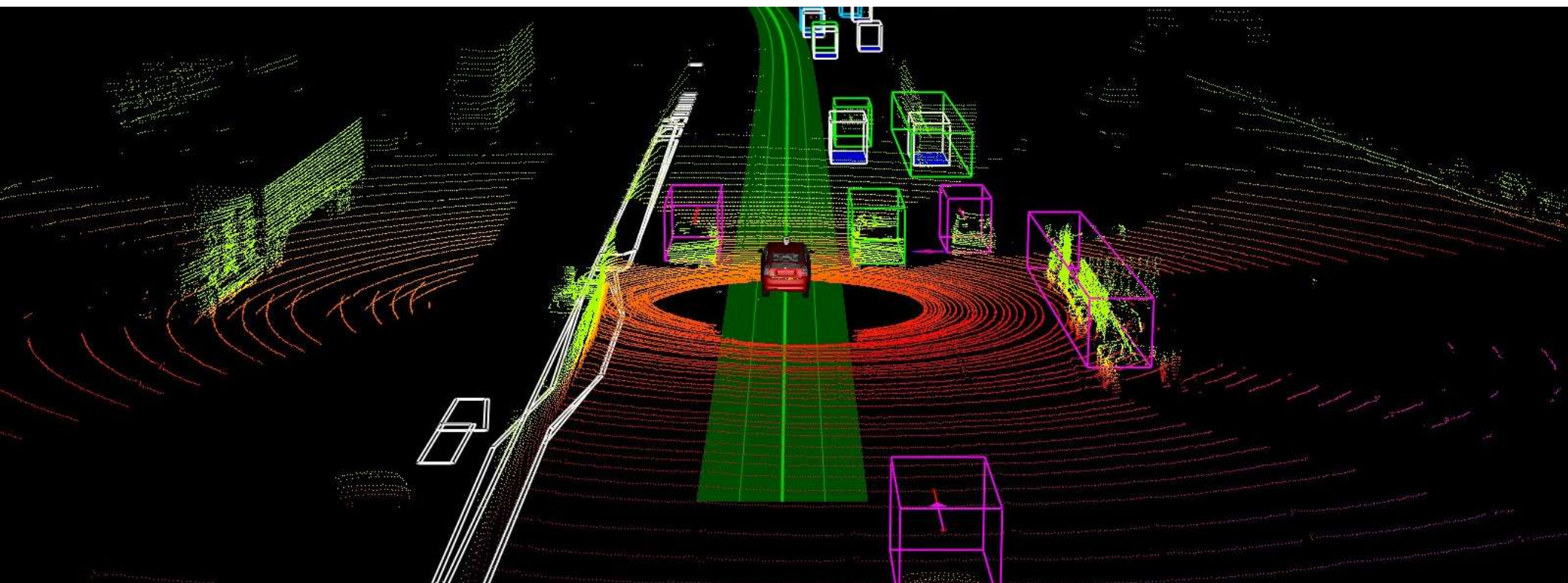




SPARSE EMBEDDED CONVOLUTIONAL DETECTION TRAINING ON CUSTOM DATA

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AGENDA

1. Data Preparation
2. Environment Setup
3. Debugging & Code Setup
4. Training Preparation
5. Training Execution
6. Results & Observations

Mentor : Ankit sir and Aniket sir
Course Instructor: Dr. Manisha Pattanaik

velodyne	Sort A to Z	me	Mar 13 me
label_2_id		Aniket P	Mar 23
label_2		Aniket P	Mar 23
image_2		me	Mar 13 me
calib_padded		Aniket P	Mar 23
calib			



DATA PREPARATION

1. Collection of Data (2000+ Images collected)
2. Filtering of Usable images
3. Finally 1118 images were chosen
4. Generated Depth analysis using midas to generate velodyne and calib for data
5. It needs lidar data point (Since no access hence we used Midas)
6. Use of yolov8m to generate labels class restricted still got 9 classes including (Car, Bicycle, Motorcycle, Bus, Auto, Animal, Pedestrian, Truck, Tractor)

ISSUE DURING DATA CREATION

Importing images from Drive and processing them to use midas generate velodyne,calib but some images content cannot be loaded.

```
[ ] for img in broken:
    path = os.path.join(raw_path, img)

    if os.path.exists(path):
        size = os.path.getsize(path)
        print(img, size)
    else:
        print(img, "MISSING")

img_427.jpg MISSING
img_428.jpg MISSING
img_432.jpg MISSING
img_433.jpg MISSING
img_435.jpg MISSING
img_437.jpg MISSING
img_438.jpg MISSING
img_439.jpg MISSING
img_441.jpg MISSING
img_442.jpg MISSING
img_443.jpg MISSING
```

- Dataset expected **continuous numbering (000000.png format)**
- But some images were missing
- This breaks KITTI-style indexing

Projected LiDAR points



ENVIRONMENT SETUP

◆ Issues faced

- Started with **MMDetection3D** on Google Colab
- Faced **mmcv** installation issues
- Encountered **PyTorch–CUDA** compatibility problems
- Repository incompatible with **latest GPU** setup

◆ Analysis & Decision

Identified **dependency conflicts** as root cause

Switched to **OpenPCDet** framework

Chosen for **modularity** and **better compatibility**

◆ Using WSL

Created **stable Linux** environment

Managed **Python** virtual environments

Ensured **CUDA–PyTorch** compatibility

Accessed **datasets** from **Windows** drives

```
Requirement already satisfied: markdown-it-py<2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich==13.8.0) (1.0.0)
Requirement already satisfied: pygments<3.0.0, >2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich==13.8.0) (2.19.2)
Requirement already satisfied: mdurl<0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py<2.2.0->rich==13.8.0) (0.1.2)
Requirement already satisfied: six<2.1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<2.7->matplotlib==3.8.0) (1.17.0)
Building wheels for collected packages: mmcv
  Building wheel for mmcv (setup.py) ... done
  Created wheel for mmcv: filename=mmcv-2.1.0-py3-none-any.whl size=719680 sha256=7a2afe3cd8d4933f5824f82df4514c32493941960b36a9cd97e9b6d1d52481
  Stored in directory: /root/.cache/pip/wheels/69/ad/f4/c99a3ff8cf71ba08d178fb3071c096b7f825274b27f8f86
Successfully built mmcv
Installing collected packages: mmcv
Successfully installed mmcv-2.1.0

# CELL 5 - Verify mmcv
import mmcv
from mmcv.ops import nms
print('mmcv OK:', mmcv.__version__)

ModuleNotFoundError: Traceback (most recent call last):
  /tmp/ipykernel_6087/4119908576.py in <cell line: 0>()
    1 # CELL 5 - Verify mmcv
    2 import mmcv
    3 from mmcv.ops import nms
    4 print('mmcv OK:', mmcv.__version__)

4 frames
/usr/local/lib/python3.12/dist-packages/mmcv/utils/dl_utils/hub.py in <module>
    126     raise error
    127 else:
    128     from torch.utils.model_zoo import load_url # type: ignore # noqa: F401
ModuleNotFoundError: No module named 'torch.utils.model_zoo'

NOTE: If your import is failing due to a missing package, you can
```

```
Successfully installed addict-2.4.0 contourpy-1.3.2 cycycler-0.12.1 fonttools-4.62.1 kiwisolver-1.5.0 matplotlib-3.10.8 mmengine-0.10.7 opencv-python-4.13.0.92 platformdirs-4.9.4 pyparsing-3.3.2 termcolor-3.3.3 toml-2.4.1 yapf-0.43.0
(second) dhruv@ANTARLOQ:~$ pip install "mmcv>=2.0.0"
Looking in links: https://download.openmmlab.com/mmcv/dist/cu128/torch2.12.0/index.html
Collecting mmcv>=2.0.0
  Downloading mmcv-2.2.0.tar.gz (479 kB)
  Installing build dependencies ... done
  Getting requirements to build wheel ... error
error: subprocess-exited-with-error

× Getting requirements to build wheel did not run successfully.
  exit code: 1
  [17 lines of output]
    Traceback (most recent call last):
      File "/home/dhruv/yes/envs/second/lib/python3.10/site-packages/pip/_vendor/pyproject_hooks/_in_process/_in_process.py", line 389, in <module>
        main()
      File "/home/dhruv/yes/envs/second/lib/python3.10/site-packages/pip/_vendor/pyproject_hooks/_in_process/_in_process.py", line 373, in main
        json_out["return_val"] = hook(**hook_input["kwargs"])
      File "/home/dhruv/yes/envs/second/lib/python3.10/site-packages/pip/_vendor/pyproject_hooks/_in_process/_in_process.py", line 143, in get_requires_for_build_wheel
        return hook(config_settings)
      File "/tmp/pip-build-env-1gsx2h2u/overlay/lib/python3.10/site-packages/setuptools/build_meta.py", line 333, in get_requires_for_build_wheel
        return self._get_build_requires(config_settings, requirements=[])
      File "/tmp/pip-build-env-1gsx2h2u/overlay/lib/python3.10/site-packages/setuptools/build_meta.py", line 301, in _get_build_requires
        self.run_setup()
      File "/tmp/pip-build-env-1gsx2h2u/overlay/lib/python3.10/site-packages/setuptools/build_meta.py", line 520, in run_setup
        super().run_setup(setup_script=setup_script)
      File "/tmp/pip-build-env-1gsx2h2u/overlay/lib/python3.10/site-packages/setuptools/build_meta.py", line 317, in run_setup
        exec(code, locals())
      File "<string>", line 5, in <module>
        ModuleNotFoundError: No module named 'pkg_resources'
    [end of output]

note: This error originates from a subprocess, and is likely not a problem with pip.
ERROR: Failed to build 'mmcv' when getting requirements to build wheel
(second) dhruv@ANTARLOQ:~$
```

```
>>> print(torch.cuda.get_device_name(0))
/home/dhruv/Desktop/second_model/venv/lib/python3.12/site-packages/torch/cuda/__init__.py:235: UserWarning:
NVIDIA GeForce RTX 5060 Laptop GPU with CUDA capability sm_120 is not compatible with the current PyTorch
installation.
The current PyTorch install supports CUDA capabilities sm_50 sm_60 sm_70 sm_75 sm_80 sm_86 sm_90.
If you want to use the NVIDIA GeForce RTX 5060 Laptop GPU with PyTorch, please check the instructions
at https://pytorch.org/get-started/locally/

warnings.warn(
NVIDIA GeForce RTX 5060 Laptop GPU
>>>
```

```
2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (2.6 kB)
Collecting nvidia-cudnn-cu12==9.20.0.48 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-cudnn-cu12/nvidia_cudnn_cu12-9.20.0.48-py3-none-
  27_x86_64.whl (657.9 MB)
    657.9/657.9 MB 618.3 kB/s eta 0:00:00
Collecting nvidia-cusparselt-cu12==0.7.1 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-cusparselt-cu12/nvidia_cusparselt_cu12-0.7.1-py
  linux2014_x86_64.whl (287.2 MB)
    287.2/287.2 MB 341.3 kB/s eta 0:00:00
Collecting nvidia-nccl-cu12==2.29.7 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-nccl-cu12/nvidia_nccl_cu12-2.29.7-py3-none-many
  86_64.whl (293.6 MB)
    293.6/293.6 MB 428.9 kB/s eta 0:00:00
Collecting nvidia-nvshmem-cu12==3.4.5 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-nvshmem-cu12/nvidia_nvshmem_cu12-3.4.5-py3-none
  14_x86_64.manylinux_2_17_x86_64.whl (139.1 MB)
    139.1/139.1 MB 462.5 kB/s eta 0:00:00
Collecting triton==3.6.0+git9844da95 (from torch)
  Downloading https://download-r2.pytorch.org/whl/nightly/triton-3.6.0%2Bg9844da95-cp312-c
  ux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cublas-cu12==12.8.4.1.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupt
  olver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)
  Downloading https://pypi.nvidia.com/nvidia-cublas-cu12/nvidia_cublas_cu12-12.8.4.1-py3-non
  2_27_x86_64.whl (594.3 MB)
    594.3/594.3 MB 271.8 kB/s eta 0:00:00
Collecting nvidia-cuda-runtime-cu12==12.8.90.* (from cuda-toolkit[cublas,cudart,cufft,cufile
  d,cusolver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)
  Downloading https://pypi.nvidia.com/nvidia-cuda-runtime-cu12/nvidia_cuda_runtime_cu12-12.8
  -manylinux2014_x86_64.manylinux_2_17_x86_64.whl (954 kB)
    954.8/954.8 kB 849.5 kB/s eta 0:00:00
Collecting nvidia-cufft-cu12==11.3.3.83.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupt
  olver,cusparselt,nvjitlink,nvrtc,nvtx]==12.8.1; platform_system == "Linux"->torch)
  Downloading https://pypi.nvidia.com/nvidia-cufft-cu12/nvidia_cufft_cu12-11.3.3.83-py3-non
  14_x86_64.manylinux_2_17_x86_64.whl (193.1 MB)
    16.0/193.1 MB 371.4 kB/s eta 0:07:57
```

1. Installed Several version of cuda and pytorch but none was compatible with Repository
2. Took 4-5 days to realize
3. Finally switch to Local GPU RTX5000
4. Moved from WSL to VS Code for better Coding Capabilities
5. Python 3.9 Finally was Used for environment setup
6. Used **PyTorch (CUDA 11.8 - cu118 build)**

DEBUGGING AND SETUP

01 KEY FILES

- **second.yaml** – Master config, all params
- **detectors/second.py** – Top-level model wiring
- **spconv_backbone.py** – 3D sparse convolution backbone
- **anchor_head_single.py** – Anchors + loss head
- **anchor_generator.py** – Builds anchor grid
- **kitti_dataset.py** – GT label loading + filter
- **setup.py** – Compiles CUDA ops

02 ANCHOR CONFIGURATION

- **anchor_sizes** [l, w, h] – Match mean GT dimensions
- **anchor_rotations** [0, $\pi/2$] – 0° and 90°
- **anchor_bottom_heights** – Align with sensor ground z
- **feature_map_stride** – Equal to backbone stride (8)
- **matched_threshold** – $\text{IoU} \geq 0.6 \rightarrow$ positive anchor
- **unmatched_threshold** – $\text{IoU} \leq 0.45 \rightarrow$ negative
- **POINT_CLOUD_RANGE** – Changes break anchor grid

03 CLASS FILTERING

- **CLASS_NAMES** in YAML – Single source of truth
- **kitti_dataset.py** – Filters GT by type string
- **Case-sensitive** (Car $\neq$ car)
- **Order must match YAML** (anchor order = class order)
- **3 classes only** $\rightarrow$ stable gradients, less noise
- **Rare classes removed** $\rightarrow$ van/misc have few samples

04 ENVIRONMENT & DEPENDENCY ERRORS (WSL2)

- **libtorch_global_deps.so error**
Fix: `export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH`
- **PIL.imaging ImportError**
Fix: `pip install --force-reinstall --no-cache-dir pillow==9.5.0`
- **pointnet2_batch_cuda ImportError**
Fix: `python setup.py develop`

06 CORRECT INSTALL ORDER & DATA PIPELINE FLOW

- **Install Order** (non-negotiable): `torch` $\rightarrow$ `numpy==1.23.5` $\rightarrow$ `scipy` $\rightarrow$ `pillow` $\rightarrow$ `scikit-image` $\rightarrow$ `llvmlite` $\rightarrow$ `numba` $\rightarrow$ `spconv-cu121` $\rightarrow$ `Kornia`
- `seconv` / `setup.py develop`

Data Pipeline: `second.yaml` $\rightarrow$ `kitti_dataset.py` $\rightarrow$ `data_processor.py` $\rightarrow$ `spconv_backbone`

05 DEBUGGING CHECKLIST

- **Loss not decreasing** $\rightarrow$ Anchors misaligned: check `bottom_heights` & `sizes`
- **mAP = 0** $\rightarrow$ Class name mismatch OR anchor order mismatch in YAML
- **NaN loss** $\rightarrow$ Zero positives (fix anchors) or LR too high
- **CUDA op import fail** $\rightarrow$ `pedet/ops/*_cuda.so` (should be 5 files)
- **OOM crash** – Reduce `BATCH_SIZE_PER_GPU` to 2; confirm stable first
- **GT boxes dropped** – `POINT_CLOUD_RANGE` too tight
- **Anchors not matching** $\rightarrow$ `kitti_dataset.py` debug logs
- **Wrong class predictions** – Check YAML class order vs: anchor config
- **Import torch fails** – Wrong NVIDIA driver vs `torch setup.py develop`

06 CORRECT INSTALL ORDER & DATA PIPELINE FLOW

`second.yaml` $\rightarrow$ `kitti_dataset.py` $\rightarrow$ `data_processor.py` $\rightarrow$ `spconv_backbone.py` $\rightarrow$ `train.py`

TRAINING PREPARATION

01 Data Transfer Setup

- Mounted **Google Drive** in WSL environment.
- Accessed dataset from mounted directory
- Copied dataset to remote GPU server: **vlsi@192.168.38.50**

02 Dataset Transfer to GPU

- Used **scp** command to transfer data:

```
scp -r /mnt/gdrive/KITTI_ORIG  
vlsi@192.168.38.50:/path/to/project/
```
- Ensured data availability on GPU system

03 Dataset Organization

- **Placed dataset** in required structure:
 - data/kitti/training/
 - ← velodyne/
 - ← label_2/
 - ← calib/
 - ← image_2/

03 Dataset Organization

- Placed dataset in required structure:

```
data/kitti/training/  
← velodyne/  
← label_2/  
← calib/
```
- Ensured data availability on GPU system

04 Path Configuration

- Updated dataset paths in:
 - **kitti_dataset.py**
 - **second.yaml**
- Ensured correct linking of:
 - point clouds
 - labels
 - calibration files

05 Data Preparation (GT Database)

- Generated ground truth database and info files using:

```
python -m pcdet.datasets.kitti.kitti_dataset  
create_kitti_infos tools/cfgs/dataset_configs/kitti_dataset.yaml
```
- 📁 Used for:
 - Faster training
 - Data augmentation (GT sampling)

06 Output Files Generated

- kitti_infos_train.pkl
- kitti_infos_val.pkl

07 Final Readiness

- Dataset verified and structure lelled correctly
- Info files generated successfully
- Ready for training with SECOND model

ERRORS AND CORRECTIONS

```
gt_database sample: 886/888
gt_database sample: 887/888
gt_database sample: 888/888
Database Tractor: 852
Database Pedestrian: 792
Database Car: 2554
Database Bicycle: 1372
Database Bus: 253
Database Motorcycle: 186
Database Truck: 6
Database Animal: 2
Database Auto: 1
-----Data preparation Done-----
```

Previously it was generated on basis of nine class classification then I changed the code for three classes only as training was not starting, so at three classes we got these results.

```
> all_classes.extend(info['annos'][name].tolist())
> unique, counts = np.unique(all_classes, return_counts=True)
> for u, c in zip(unique, counts):
>     print(f'{u}: {c} instances')
> "
Car: 3406 instances
Cyclist: 1558 instances
Pedestrian: 792 instances
(openpcdet env) vlsi@vlsi-Precision-5820-Tower:/media/vlsi/7CB0-FA
```

```
1 DATASET: 'kittiDataset'
2 DATA_PATH: '/media/vlsi/7CB0-FACC/OpenPCDet_custom/data/kitti'
3 POINT_CLOUD_RANGE: [0, -40, -3, 70.4, 40, 1]
4
5 DATA_SPLIT: {
6     'train': train,
7     'test': val
8 }
9
10 TRAIN_DATA_PATH:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
(openpcdet env) vlsi@vlsi-Precision-5820-Tower:/media/vlsi/7CB0-FACC/OpenPCDet_custom/tools$ python train.py \
--cfg_file cfgs/custom_models/second.yaml \
--epochs 10
326-04-07 10:33:11,682 INFO cfg.OPTIMIZATION.LR_CLIP: 1e-07
326-04-07 10:33:11,682 INFO cfg.OPTIMIZATION.LR_WARMUP: False
326-04-07 10:33:11,682 INFO cfg.OPTIMIZATION.WARMUP_EPOCH: 1
326-04-07 10:33:11,682 INFO cfg.OPTIMIZATION.GRAD_NORM_CLIP: 10
326-04-07 10:33:11,682 INFO cfg.TAG: second
326-04-07 10:33:11,682 INFO cfg.EXP_GROUP_PATH: custom_models
326-04-07 10:33:11,687 INFO ----- Create dataloader & network & optimizer -----
326-04-07 10:33:11,688 INFO Loading KITTI dataset
326-04-07 10:33:11,688 INFO Total samples for KITTI dataset: 0
Traceback (most recent call last):
  File "/media/vlsi/7CB0-FACC/OpenPCDet_custom/tools/train.py", line 209, in <module>
    main()
  File "/media/vlsi/7CB0-FACC/OpenPCDet_custom/tools/train.py", line 110, in main
    train_set, train_loader, train_sampler = build_dataloader(
  File "/media/vlsi/7CB0-FACC/OpenPCDet_custom/tools/./pcdet/datasets/__init__.py", line 87, in build_dataloader
    dataloader = DataLoader(
  File "/opt/openpcdet_envs/openpcdet_env/lib/python3.9/site-packages/torch/utils/data/dataloader.py", line 266, in __init__
    sampler = RandomSampler(dataset, generator=generator) # type: ignore
  File "/opt/openpcdet_envs/openpcdet_env/lib/python3.9/site-packages/torch/utils/data/sampler.py", line 103, in __init__
    raise ValueError("num_samples should be a positive integer "
ValueError: num_samples should be a positive integer value, but got num_samples=0
```

TRAINING EXECUTION

01 Training Command

- Started training using:

```
python tools/train.py --cfg_file
tools/cfgs/custom_models/second.yaml
```

02 Model Initialization

- Loaded configuration from second.yaml
- Built dataset and SECOND model
- Initialized weights and optimizer

03 GPU Utilization

- Training executed on **remote GPU** (vlsi server)
- Verified CUDA and memory usage
- Adjusted batch size for stability

04 Training Process

- Forward pass → Loss computation → **Backpropagation**
- Anchor matching and target assignment
- Iterative weight updates

05 Monitoring

- Observed **loss decreasing trend**
- Checked logs for **errors and stability**

06 Output Generated

- Model checkpoints (.pth) saved
- Logs stored for analysis

Per-class prediction summary:

Car: GT=459, Predicted=1091

Pedestrian: GT=101, Predicted=0

Cyclist: GT=183, Predicted=1

(openpcdet_env) vlsi@vlsi-Precision-5820-Tower

```
python tools/train.py --cfg_file cfgs/kitti_models/second.yaml \
--batch_size 1 --workers 0 \
--epochs 40 \
--num_epochs_to_eval 0
2026-04-11 22:49:13.353 INFO Train: 21/40 ( 52%) [ 819/874 ( 94%)] Loss: 6.381 (6.63) LR: 2.698e-03 Time cost: 02:18/00:09 [35:18/46:59] Acc_iter 18300
Data time: 0.03(0.03) Forward time: 0.30(0.14) Batch time: 0.33(0.17)
/bin/sh: 1: gnuaste: not found
2026-04-11 22:49:13.434 INFO
2026-04-11 22:49:25.165 INFO Train: 21/40 ( 52%) [ 869/874 ( 99%)] Loss: 6.772 (6.58) LR: 2.691e-03 Time cost: 02:30/00:00 [35:30/47:54] Acc_iter 18350
Data time: 0.01(0.03) Forward time: 0.09(0.15) Batch time: 0.10(0.17)
2026-04-11 22:49:25.956 INFO Train: 21/40 ( 52%) [ 873/874 (100%)] Loss: 14.62 (6.59) LR: 2.690e-03 Time cost: 02:31/00:00 [35:31/47:55] Acc_iter 18354
Data time: 0.01(0.03) Forward time: 0.09(0.15) Batch time: 0.10(0.17)
epochs: 374 |
2026-04-11 22:49:26.352 INFO Train: 22/40 ( 55%) [ 0/874 ( 0%)] Loss: 4.321 (4.32) LR: 2.690e-03 Time cost: 00:00/04:42 [35:31/1:29:32] Acc_iter 1835
Data time: 0.01(0.01) Forward time: 0.31(0.31) Batch time: 0.32(0.32)
2026-04-11 22:49:36.444 INFO Train: 22/40 ( 55%) [ 45/874 ( 5%)] Loss: 2.526 (7.01) LR: 2.684e-03 Time cost: 00:10/03:07 [35:41/1:02:29] Acc_iter 184
Data time: 0.04(0.02) Forward time: 0.40(0.20) Batch time: 0.45(0.23)
2026-04-11 22:49:47.501 INFO Train: 22/40 ( 55%) [ 95/874 (11%)] Loss: 3.166 (6.51) LR: 2.677e-03 Time cost: 00:21/02:54 [35:52/1:01:33] Acc_iter 184
Data time: 0.02(0.02) Forward time: 0.14(0.20) Batch time: 0.15(0.22)
/bin/sh: 1: gnuaste: not found
2026-04-11 22:49:47.575 INFO
```



"I executed training using train.py with the configured SECOND model, monitored loss and GPU usage, and generated model checkpoints successfully."

RESULTS AND OUTCOMES

```

2026-04-12 05:40:18,616 INFO ==> Done (loaded 163/163)
2026-04-12 05:40:18,619 INFO ***** EPOCH 40 EVALUATION *****
eval: 100%|██████████████████████████████████████████████████████████| 106/106 [00:12<00:00, 8.36it/s, recall_0.3=(0, 1) / 743]
2026-04-12 05:40:31,296 INFO ***** Performance of EPOCH 40 *****
2026-04-12 05:40:31,296 INFO Generate label finished(sec_per_example: 0.1196 second).
2026-04-12 05:40:31,296 INFO recall_roi_0.3: 0.000000
2026-04-12 05:40:31,296 INFO recall_rcnn_0.3: 0.001346
2026-04-12 05:40:31,297 INFO recall_roi_0.5: 0.000000
2026-04-12 05:40:31,297 INFO recall_rcnn_0.5: 0.000000
2026-04-12 05:40:31,297 INFO recall_roi_0.7: 0.000000
2026-04-12 05:40:31,297 INFO recall_rcnn_0.7: 0.000000
2026-04-12 05:40:31,297 INFO Average predicted number of objects(106 samples): 10.302

```

1. Training Status

Model trained successfully up to **Epoch 40**

Evaluation completed on validation dataset

Total samples processed: **106/106**

2. Performance Metrics

- Recall (ROI, IoU=0.3): 0.0000

- Recall (RCNN, IoU=0.3): 0.0013

- Recall (IoU=0.5 & 0.7): 0.0000

3. Prediction Statistics

- Average predicted objects per sample:
10.302

- Label generation time: **~0.12 sec/sample**

“The model completed training up to 40 epochs, but evaluation showed very low recall, indicating issues with detection performance and the need for further tuning and update of data (labels/calib).”

SECOND Model in OpenPCDet (Repo-Based Explanation)

Point Cloud → Voxelization → VFE → Sparse Conv → BEV → 2D CNN → Detection Head

